

AWS EC2 COMPREHENSIVE CONFIGURATION STUDY GUIDE

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Domain: AWS Core Compute Infrastructure

1. What is Amazon EC2?

Amazon Elastic Compute Cloud (Amazon EC2) is a web service that provides secure, resizable compute capacity in the cloud. It is designed to make web-scale cloud computing easier for developers by allowing them to obtain and configure virtual servers (termed "instances") with minimal friction. EC2 changes the economics of computing by allowing you to pay only for capacity that you actually use.

2. Core Architectural Pillars: Availability, Scalability, & Redundancy



- **Availability (High Availability):** Achieved by deploying EC2 instances across multiple Availability Zones (AZs). If an underlying data center failure disrupts an entire AZ, instances in surviving zones continue serving traffic without application downtime.
- **Scalability:** EC2 offers two methods of scaling:
 - *Vertical Scaling (Scaling Up/Down):* Increasing or decreasing the instance size (e.g., upgrading from a `t3.micro` to an `m5.large`) to provide more vCPU and memory to a single system.
 - *Horizontal Scaling (Scaling Out/In):* Adding or removing identical EC2 instances to share the workload, typically orchestrated by **AWS Auto Scaling** and distributed by an Elastic Load Balancer (ELB).
- **Redundancy:** Eliminates single points of failure by ensuring that secondary or standby EC2 instances, replica storage volumes, and isolated network paths exist across different fault domains.

3. Key Features of Amazon EC2

- **Virtual Computing Environments:** Known as instances, which can be custom configured with vCPU, Memory, Storage, and Networking capacity.
- **Preconfigured Templates:** Uses Amazon Machine Images (AMIs) to package the OS, application configuration, and software dependencies required for instantaneous deployments.

- **Flexible Pricing Models:** Options include On-Demand, Savings Plans, Reserved Instances (RI), and Spot Instances for optimizing infrastructure costs.
- **Secure Access:** Integrates tightly with AWS Identity and Access Management (IAM) and uses Key Pairs for secure remote login authentication.

4. Amazon Machine Image (AMI) Details

An AMI provides the complete software blueprint for your instance. It includes a root volume template (Operating System, tools, and app software), launch permission tokens, and block device mappings specifying the storage layout upon boot.

Processor Architecture Types:

- **x86_64 (Intel / AMD):** Standard corporate architecture ensuring legacy framework compatibility.
- **arm64 (AWS Graviton):** Highly efficient, custom-built AWS silicon providing up to 40% higher price-performance efficiency over comparable x86 variants.

5. Storage: Amazon EBS vs. Instance Store

VISUAL MAPPING: EBS (NETWORK) VS INSTANCE STORE (LOCAL HARDWARE)

Amazon EBS Volume

Network-connected via network fabric.
Data persists beyond stop/termination.

EC2 Instance Store

Physically attached to the host server hardware.
Ephemeral: Data lost if stopped/fails.

Feature	Amazon EBS (Elastic Block Store)	Instance Store
Storage Type	Network-attached block storage.	Physically attached to the underlying host server hardware.
Persistence	Data survives instance stop, restart, and failure.	Ephemeral: Data is completely lost if the instance stops, terminates, or encounters underlying hardware failure. Survives basic reboots.
Performance	Sub-millisecond latencies, scalable IOPS up to massive limits via network interface.	Ultra-low latencies, high random I/O performance, and massive throughput (ideal for scratch spaces and caches).
Backup capabilities	Supports point-in-time snapshots to Amazon S3 seamlessly.	Must be manually backed up at the software or application layer.

6. Networking: ENI vs. ENA vs. EFA

Interface Type	Description & Performance Focus	Primary Use Case
ENI (Elastic Network Interface)	A logical networking component in a VPC representing a virtual network card. Includes primary/secondary private IPs, Mac addresses, and Security Groups.	Standard business production web applications, management networks, multi-homed configurations.
ENA (Elastic Network Adapter)	Provides Enhanced Networking capabilities. Engineered to deliver higher bandwidth (up to 100+ Gbps) and lower CPU utilization.	High-throughput web setups, large relational databases, big data pipelines.
EFA (Elastic Fabric Adapter)	An ENA built with single root I/O virtualization (SR-IOV) and added OS-bypass capabilities, bypassing the operating system kernel to communicate directly with cluster hardware nodes. High performance, ultra-low latency.	High-Performance Computing (HPC), Machine Learning distributed training models, complex fluid dynamics simulation clusters.

7. Instance Family Types and Sizes

EC2 instances follow an explicit alphanumeric naming convention (e.g., `c6i.xlarge`) and are segmented into core optimized archetypes:

- **General Purpose (T, M):** Balanced distributions of compute, memory, and networking capacity.
- **Compute Optimized (C):** Tailored with high-performance processors for compute-heavy batch processing and encoding jobs.
- **Memory Optimized (R, X):** Tailored for memory-resident architectures processing massive datasets (e.g., Redis clusters).
- **Storage Optimized (I, D, H):** Maximized sequential read/write structures suited for NoSQL and distributed cluster file systems.
- **Accelerated Computing (P, G, Inf):** Infused with physical GPU coprocessors for AI training and graphics rendering pipelines.

8. Network Configuration & Security Groups

- **Subnet Topologies:** System placement dictates whether an instance is reachable from the internet (Public Subnet via an Internet Gateway) or isolated within internal corporate lines (Private Subnet relying on an outbound NAT Gateway).
- **Security Groups:** Acts as a **stateful virtual firewall** applied directly around the instance level. Security groups evaluate traffic filters conditionally:
 - Rules are *permissive only* (you cannot define restrictive deny rules; everything unlisted is explicitly dropped).
 - Because they are *stateful*, outbound responses to allowed inbound requests are permitted to travel outward automatically.

9. Comprehensive Amazon EBS Volume Types

Volume Category	Volume Type API Name	Max IOPS / Volume	Max Throughput	Typical Target Workload
SSD-backed Volumes	gp3 (General Purpose SSD)	16,000 IOPS	1,000 MiB/s	System boot volumes, normal virtual desktops, low-latency applications. Balanced price-performance.
	gp2 (General Purpose SSD)	16,000 IOPS	250 MiB/s	Older generation general-purpose volume; performance scales linearly with size.
	io2 / io2 Block Express (Provisioned IOPS)	256,000 IOPS	4,000 MiB/s	Mission-critical business databases (Oracle, SQL Server), high-capacity transaction processing.
HDD-backed Volumes	st1 (Throughput Optimized HDD)	500 IOPS	500 MiB/s	Frequently accessed, throughput-intensive workloads, MapReduce cluster nodes, large log arrays.
	sc1 (Cold HDD)	250 IOPS	250 MiB/s	Infrequently accessed data, archiving applications needing cheap storage space. Lowest cost.